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Netara: A Decentralized AI Inference Framework for Circular Compute via Idle Edge Devices

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NETARA | SCAN FOR MORE

01 INTRODUCTION

AI workloads are expanding rapidly, deepening reliance on centralised data centres and driving heavy energy and infrastructure demand. Meanwhile, billions of edge devices run at low average utilisation.

Netara applies circular economy principles to computing: idle capacity becomes a recoverable resource, turning existing devices into a distributed AI inference network — no new infrastructure.

02 OBJECTIVES

- ✓ Turn idle edge devices into a decentralised compute network.
- ✓ Build an idle-time device agent and micro-task orchestrator.
- ✓ Verify results across heterogeneous nodes via consensus.
- ✓ Project cost and energy savings vs centralised inference.

03 METHODOLOGY

Netara has four components: an idle-time device agent; an orchestrator that splits jobs into micro-tasks; consensus verification via overlapping assignment; and a REST interface.

Feasibility was assessed on a prototype of heterogeneous devices running batch NLP inference; projections vs centralised cloud costs.

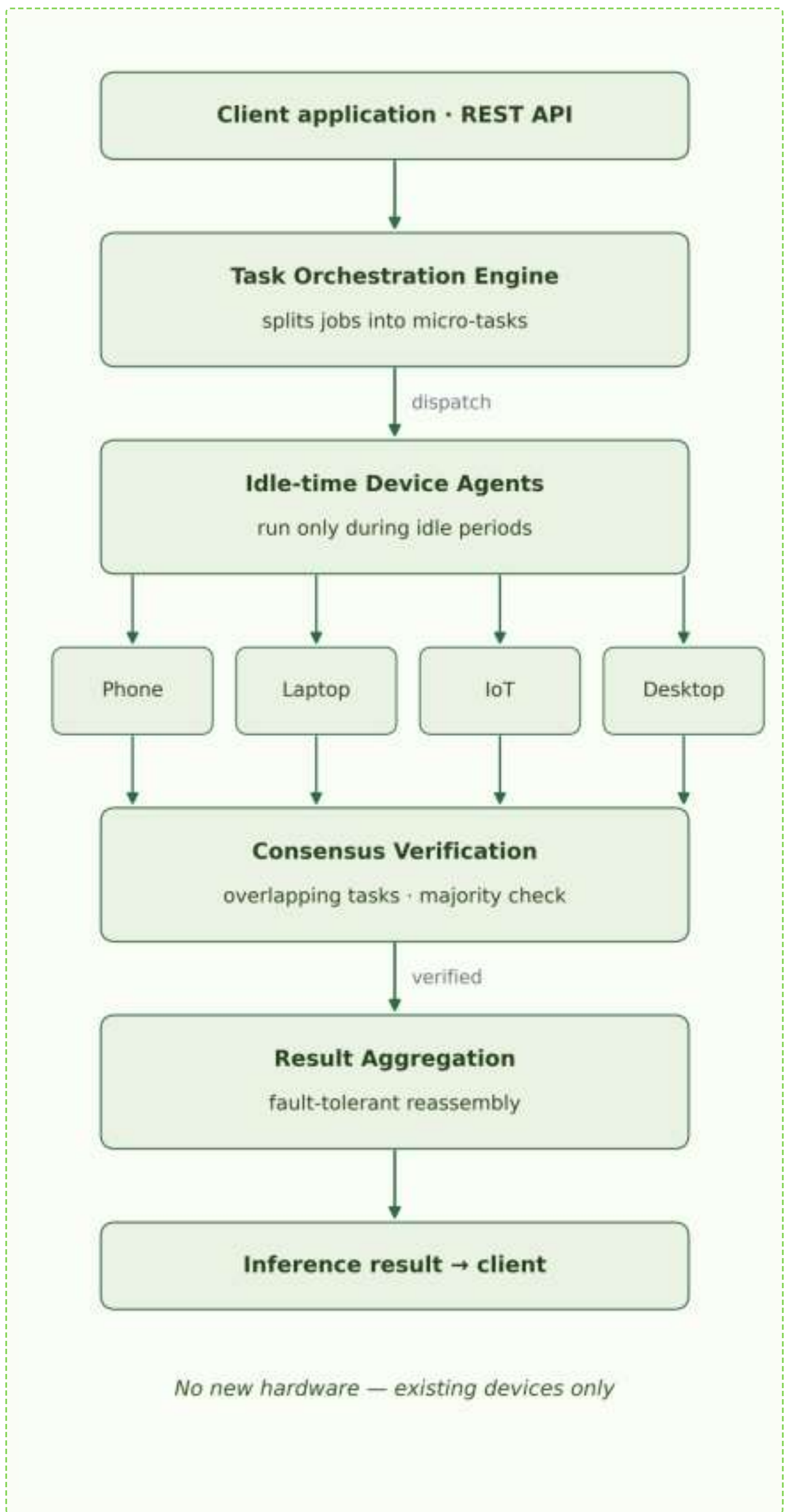

Figure 1. Netara system architecture.

04 RESULTS

The prototype demonstrates parallel micro-task execution and fault-tolerant inference across heterogeneous nodes. Projected infrastructure cost reduction of 60–80%.

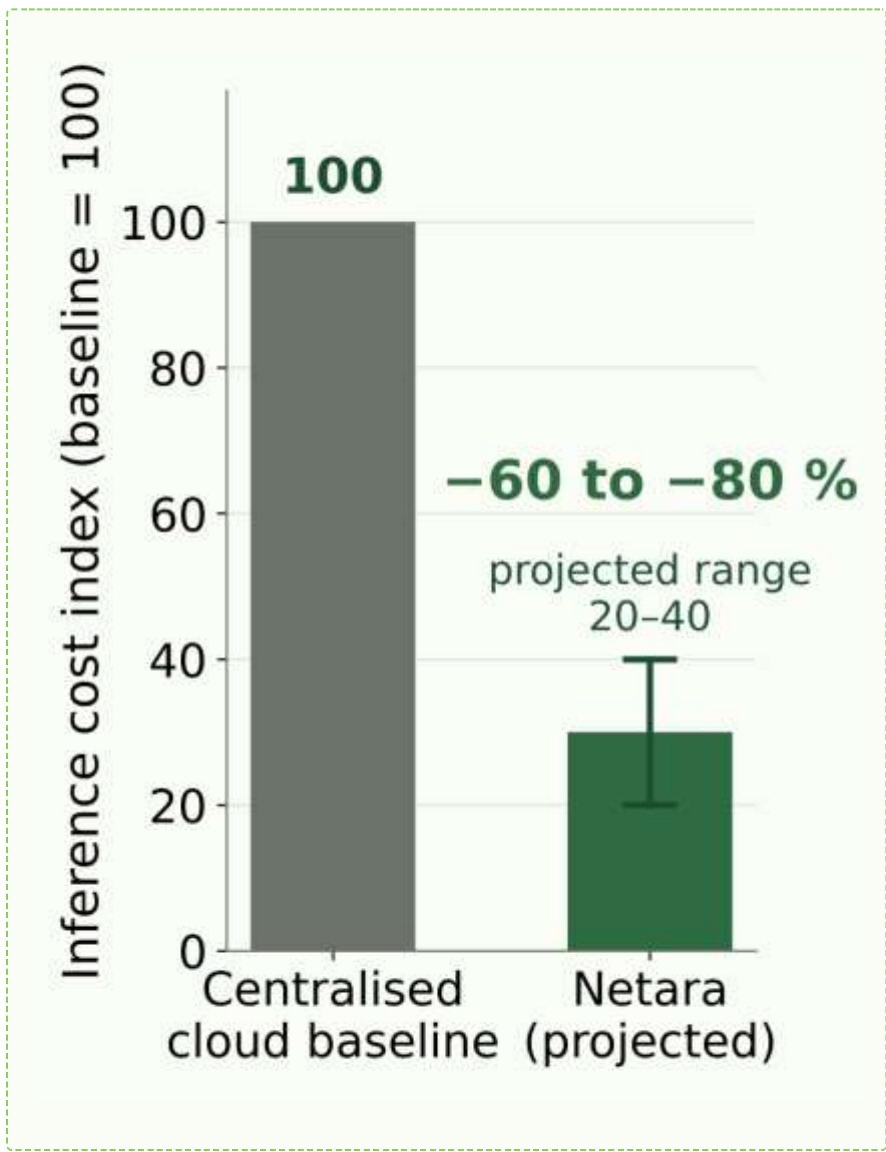

Figure 2. Projected cost index (baseline = 100).

PARAMETER	BASELINE	THIS WORK	Δ %
Parallel micro-tasks	N/A	Yes	—
Fault-tolerant inference	N/A	Yes	—
Cost index (projected)	**100**	**20–40**	**−60–80**
New hardware needed	Servers	None	—

Table 1. Prototype outcomes vs baseline.

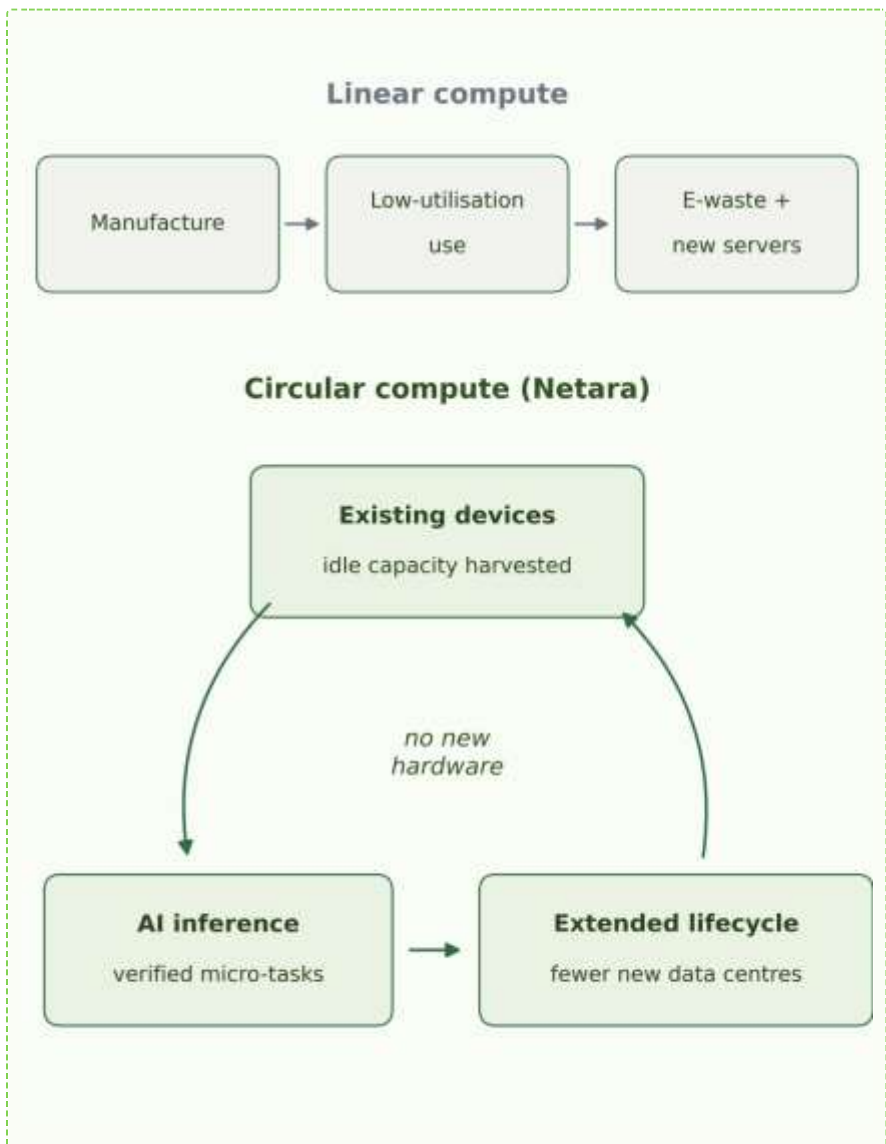

Figure 3. Linear vs circular compute lifecycle.

05 DISCUSSION

Batch inference runs reliably on idle devices — volunteer computing plus consensus verification. Analytical projections; large-scale validation pending. Challenges: heterogeneity, incentives, privacy.

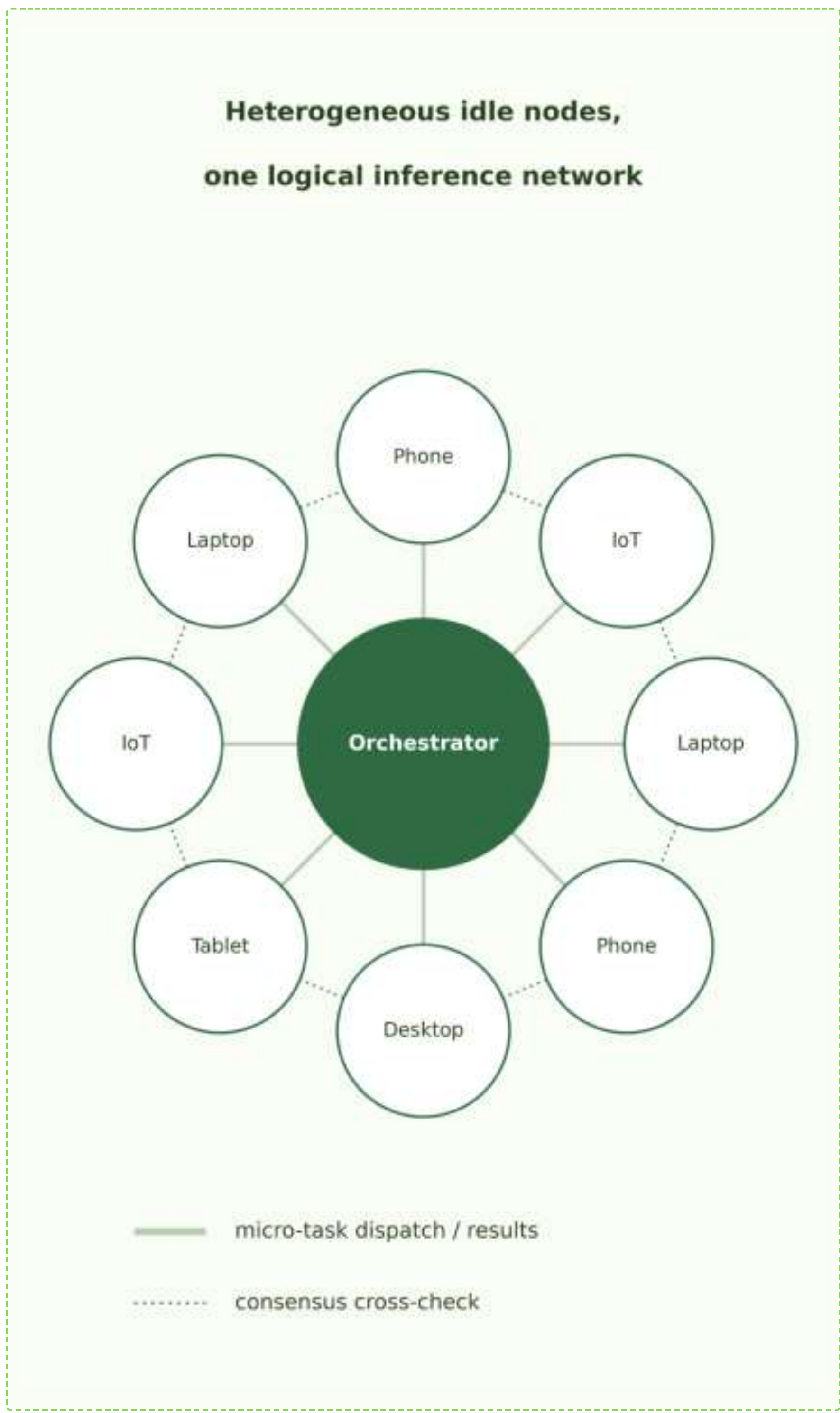

Figure 4. Prototype network topology and cross-checks.

06 CONCLUSIONS

- Fault-tolerant inference on idle edge devices is feasible.
- Projected infrastructure cost reduction of 60–80%.
- Circular reuse: affordable AI for the Global South.

> "Idle devices are latent AI infrastructure — 60–80% lower cost.
>
> — KEY FINDING

07 FUTURE WORK

Next: a pilot deployment on 10 heterogeneous edge devices with a Python orchestrator; a 1,000-task DistilBERT benchmark vs a centralised cloud baseline; measured energy accounting; and incentive and privacy mechanisms for open participation.

08 REFERENCES

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